Minimum Time Difference

0 1

["23:59", "00:00"]

ex → | 05:45 | | | | | | | | | |

↳ 5x60+minutes = 345 m

① Convert all into minutes

730	615	795	1040	1080	1187	1439

② find minimum difference

730	615	795	1040	1080	1187	1439

Compare

Compare

$O(n^2)$

for this → first sort all (nlogn)

615	730	795	1040	1080	1187	1439

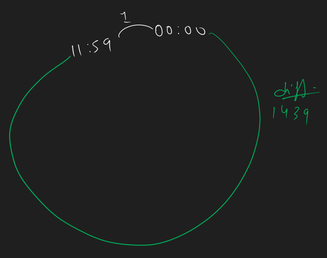
a) → Compare Compare Compare Compare Compare Compare

b) → min

③ last case — Jaha fass skte hai

11:59 ——— 00:00 → this will give 1439

24:00 ⇒ 1440 min



⇒ lastdiff = (minutes[0] + 1440) − minutes[n−1];

min = (min, lastDiff)

vector<int> minutes;

① for (i=0 ⟶ n)

{

string currTime = timepoint[i];

hour = stoi(currTime.substr(0,2));

minute = stoi(" .substr(3,2));

TotalTime = hour * 60 + minute;

minutes.push_back(TotalTime);

}

find min ⇒

Sort(minutes.begin(), minutes.end())

mini = INT_MAX

for (i=0 ⟶ minutes.size() − 1)

{

diff = minutes[i+1] − minutes[i];

mini = min (mini, diff);

}

24:00 ⇒ 00:00 + 1440 — 23:59 = 1

edge case ⇒ Important!

LastDiff = (minutes[0] + 1440) — minutes[n−1]

mini = min (mini, LastDiff);

return mini